Choose a motion model before an equation

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Constant acceleration / ES: aceleración constante / PT: aceleração constante

Velocity changes by the same amount each second.

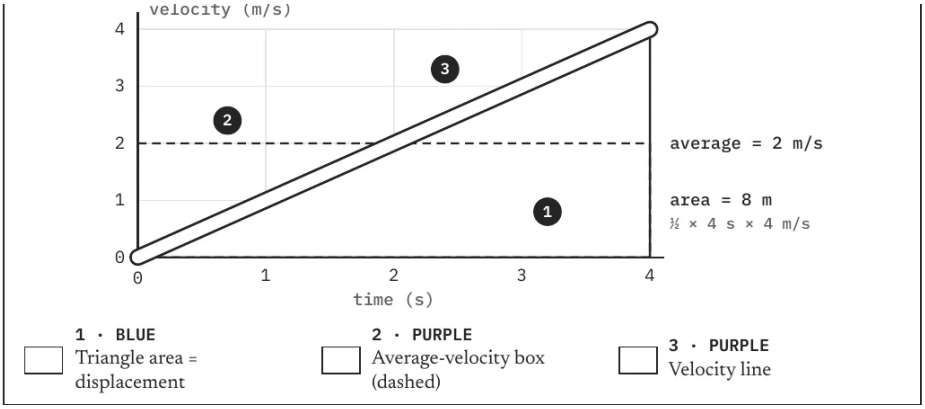
Also on your note page: Steady velocity, Average velocity, Model selection.

B READ THE RELATIONSHIP

Constant acceleration means equal changes in velocity in equal times.

Choose the equation by the given and missing quantities

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement.

Given $v_i = 0$, $v_f = 8$ m/s $t = 4$ s constant a

Rule $\Delta x = \frac{1}{2}(v_i + v_f) \times t$

Work average $= (0 + 8) \div 2 = 4$ m/s $\Delta x = 4 \times 4$

Answer +16 m average works: a is constant

E SAVE FOR THE FINAL MEETING

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

Read the physics: Constant acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

List what you know and what you want. A steady velocity needs no acceleration. A steady change in velocity needs a constant-acceleration model.

02 HOW IT WORKS

On a velocity–time graph, slope gives acceleration. The signed area gives displacement. For constant acceleration, the average of the first and last velocities gives the average velocity.

03 WHAT TO CHECK

Choose an equation that uses your known values. Then check the sign and the unit. A matching graph gives a second way to test the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement. +16 m average works: a is constant.

05 CONNECT TO XJ

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

SAY IT BEFORE YOU WRITE

What must stay constant to average the endpoint velocities? I chose ___ because I know ___ and need ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Constant acceleration / ES: aceleración constante / PT: aceleração constante

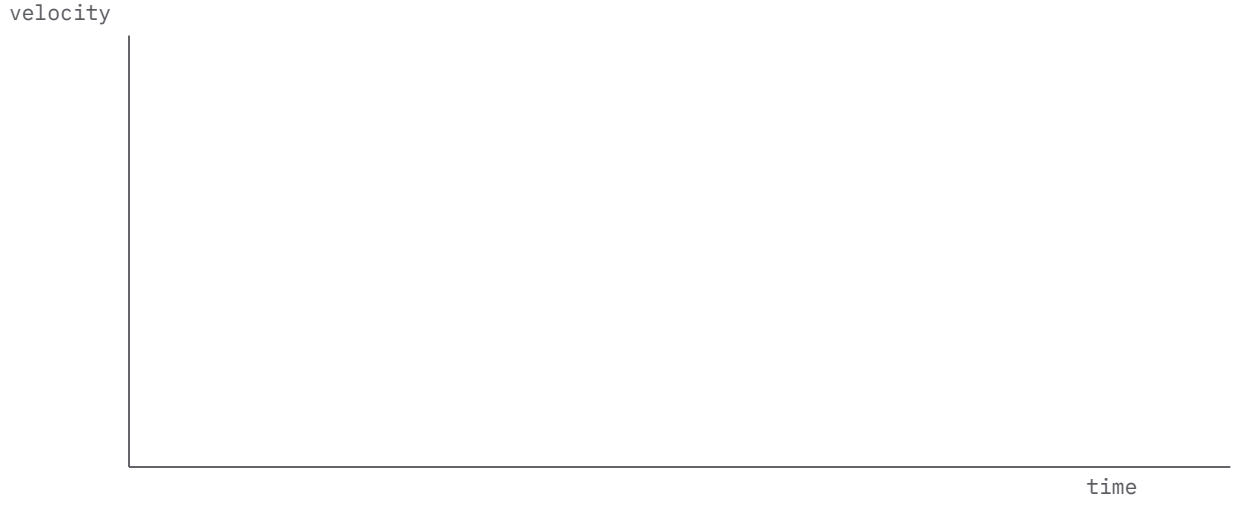
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, constant acceleration means …	The velocities are 0, 2 and 4 m/s, one second apart. What repeats? Each second, the velocity changes by ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.

Labels you can use: given · wanted · constant.



Tell a partner: I chose ___ because I know ___ and need ___.

On assignment: Constant acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

XJ moves from +2 to +6 km/s over 2 s. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?

PICK A JOB / DELIVER ONE PRODUCT

☐ **Camera engineer**

Draw the velocity–time trace and shade the area. Send the camera its displacement for the end of the shot.

☐ **Replay editor**

Sketch the opening and closing speeds and give a voice-over explaining why using the final speed for the whole clip would miss the shot.

Word bank: given · wanted · constant.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”